Section 02: Mathematical Ability

Module 15: YouTube Video Lectures & Marathon PYQ Master Vault (Infosys Systems Engineer)

> **Document Metadata & Provenance:** - **Source Corpus:** Authentic Infosys Systems Engineer (SE) YouTube Video Lectures, Live Slot Simulations & Assessment Analyses. - **Primary Channels:** OnlineStudy4U (Pratik Sir), CampusMonk (Rachit Sir), KG Placement (Yash Jain Sir), CareerRide. - **Curriculum Target:** Section 02: Mathematical Ability & Data Interpretation (Tables, Bar Graphs, Pie Charts, Pipes & Cisterns, Speed Time & Distance, Probability, Number System). - **Total Questions:** 45 Forensic Questions with complete step-by-step mathematical proofs, algebraic equations, and KaTeX derivations. - **Zero Placeholders:** Complete dataset tables, formula applications, and shortcut tricks.

Question 1

Source: OnlineStudy4U — *Infosys & LTM Aptitude Mock Test | Data Interpretation (Pratik Sir)* The following table shows the export value (in Crore INR) of four companies (A, B, C, D) in the years 2024 and 2025:

Company	2024 Export (₹ Cr)	2025 Export (₹ Cr)
A	100	130
B	120	150
C	150	180
D	80	110

Which company recorded the highest percentage growth in exports from 2024 to 2025? - [A] Company A - [B] Company B - [C] Company C - [D] Company D

Correct Answer: [D] Company D

Step-by-Step Derivation & Mathematical Proof: 1. **Growth Formula:** $$ ext{Percentage Growth} = rac{ ext{Export}*{2025}* - *ext{Export}*{2024}}{ ext{Export}_{2024}} imes 100\%$$ 2. **Calculate for each company:** - **Company A:** $$ ext{Growth} **= rac{13**- 100}{100} imes 100\% = rac{30}{100} imes 100\% = 30.00\%$$ - **Company B:** $$ ext{Growth} **= rac{15**- 120}{120} imes 100\% = rac{30}{120} imes 100\% = rac{1}{4} imes 100\% = 25.00\%$$ - **Company C:*** $$ ext{Growth} **= rac{180 - 150}**50} imes 100\% = rac{30}{150} imes 100\% = rac{1}{5} imes 100\% = 20.00\%$$ - **Company D:** $$ ext{Growth}_D = rac{**0 - 80}{80} imes 100\% = rac{30}{80} imes 100\% = rac{3}{8} imes 100\% = 37.50\%$$ 3. **Compariso**** $$37.50\% (ext{D}) > 30.00\% (ext{A}) > 25.00\% (ext{B}) > 20.00\% (ext{C})$$ 4. **Key Exam Shortcut:** - Notice that the absolute increase for all four companies is identical: exactly $₹30 ext{ Crore}$! - When absolute increase ΔX is constant, percentage increase $rac{\Delta X}{X_0}$ is strictly **maximized when the base X_0 is minimized**! - Since Company D has the smallest initial base (80), its percentage growth is mathematically guaranteed to be the highest. 5. Therefore, **Company D** recorded the highest percentage growth (37.5%).

Question 2

Source: OnlineStudy4U — *Infosys 1SHOT Aptitude (Pratik Sir)* Two pipes, A and B, can fill a cistern when operating together in $1↑rac{7}{8}$ hours (i.e., $↑rac{15}{8}$ hours). If opened separately, Pipe B takes 2 hours more than Pipe A to fill the cistern. Find the time taken by Pipe A alone to fill the cistern. - [A] 3 hours - [B] 4 hours - [C] 5 hours - [D] 6 hours

Correct Answer: [A] 3 hours

Step-by-Step Derivation & Mathematical Proof: 1. **Define Variables:** - Let the time taken by Pipe A alone be x hours. - Then Pipe B takes $(x + 2)$ hours. 2. **Work-Rate Formulation:** $$ ext{Rate of A} = ↑rac{1}{x}, \quad ext{Rate of B} = ↑rac{1}{x + 2}$$ $$ ext{Combined Rate} = ↑rac{1}{x} + ↑rac{1}{x + 2} = ↑rac{1}{15/8} = ↑rac{8}{15}$$ 3. **Algebraic Simplification:** $$↑rac{(x + 2) + x}{x(x + 2)} = ↑rac{8}{15}$$ $$↑rac{2x + 2}{x^2 + 2x} = ↑rac{8}{15} \implies ↑rac{2(x + 1)}{x(x + 2)} = ↑rac{8}{15}$$ $$↑rac{x + 1}{x(x + 2)} = ↑rac{4}{15}$$ $$15(x + 1) = 4(x^2 + 2x)$$ $$15x + 15 = 4x^2 + 8x$$ $$4x^2 - 7x - 15 = 0$$ 4. **Solve the Quadratic Equation:** $$\Delta = (-7)^2 - 4(4)(-15) = 49 + 240 = 289 = 17^2$$ $$x = ↑rac{7 \pm 17}{2 imes 4} = ↑rac{7 \pm 17}{8}$$ - Positive root (since time cannot be negative): $$x = ↑rac{7 + 17}{8} = ↑rac{24}{8} = 3 ext{ hours}$$ 5. **Verification:** - A takes 3 hours (Rate = $↑rac{1}{3}$). - B takes $3 + 2 = 5$ hours (Rate = $↑rac{1}{5}$). - Combined rate: $↑rac{1}{3} + ↑rac{1}{5} = ↑rac{5 + 3}{15} = ↑rac{8}{15}$. - Combined time: $↑rac{15}{8} = 1.875 ext{ hours} = 1 ext{ hr } 52.5 ext{ min}$. Exact match! 6. Pipe A alone takes **3 hours**.

Question 3

Source: OnlineStudy4U — *Infosys System Engineer PYQ Most Important Topics (Pratik Sir)* A bag contains 5 red balls, 4 green balls, and 3 blue balls. If two balls are drawn at random without replacement, what is the probability that both balls are of different colors? - [A] $↑rac{47}{66}$ - [B] $↑rac{19}{66}$ - [C] $↑rac{23}{33}$ - [D] $↑rac{31}{66}$

Correct Answer: [A] $↑rac{47}{66}$

Step-by-Step Derivation & Complement Rule: 1. **Total Number of Balls:** $$N = 5 + 4 + 3 = 12 ext{ balls}$$ 2. **Total Possible Outcomes:** Number of ways to choose 2 balls out of 12: $$□inom{12}{2} = □rac{12 imes 11}{2} = 66$$ 3. **Use the Complement Rule:** $$\mathbf{P}(ext{Both Different}) = 1 - \mathbf{P}(ext{Both Same Color})$$ - Both Red: $$□inom{5}{2} = □rac{5 imes 4}{2} = 10$$ - Both Green: $$□inom{4}{2} = □rac{4 imes 3}{2} = 6$$ - Both Blue: $$□inom{3}{2} = □rac{3 imes 2}{2} = 3$$ - Total ways to draw 2 balls of the same color: $$ ext{Same Color Ways} = 10 + 6 + 3 = 19$$ 4. **Probability Calculation:** $$\mathbf{P}(ext{Both Same Color}) = ↑rac{19}{66}$$ $$\mathbf{P}(ext{Both Different}) = 1 - ↑rac{19}{66} = ↑rac{66 - 19}{66} = ↑rac{47}{66}$$ 5. Result is $↑rac{47}{66}$ [A].

Question 4

Source: KG Placement — *1.4: Infosys Aptitude - Speed Time & Distance (Yash Jain Sir)* A train 240 meters long crosses a platform of length 360 meters in 30 seconds. What is the speed of the train in km/h? - [A] 64 km/h - [B] 72 km/h - [C] 80 km/h - [D] 90 km/h

Correct Answer: [B] 72 km/h

Step-by-Step Derivation & Unit Conversion: 1. **Total Distance Traversed:** When a train completely crosses a platform, the total distance covered equals the sum of the lengths of the train and the platform: $$D = L_{ ext{train}} + L_{ ext{platform}} = 240 ext{ m} + 360 ext{ m} = 600 ext{ meters}$$ 2. **Speed in m/s:** $$v = rac{D}{t} = rac{600 ext{ m}}{30 ext{ s}} = 20 ext{ m/s}$$ 3. **Convert m/s to km/h:** Multiply by the conversion factor $rac{18}{5}$: $$v_{ ext{km/h}} = 20 imes rac{18}{5} = 4 imes 18 = 72 ext{ km/h}$$ 4. Printed answer is **72 km/h**.

Question 5

Source: KG Placement — *1.13: Infosys Aptitude - Boats & Stream (Yash Jain Sir)* A boat can travel 36 km downstream in 3 hours, and 24 km upstream in 4 hours. What is the speed of the stream in km/h? - [A] 2 km/h - [B] 3 km/h - [C] 4 km/h - [D] 6 km/h

Correct Answer: [B] 3 km/h

Step-by-Step Derivation & Relative Speed: 1. Let the speed of the boat in still water be u km/h, and the speed of the stream be v km/h. 2. **Downstream Speed ($u + v$):** $$v_d = rac{36 ext{ km}}{3 ext{ h}} = 12 ext{ km/h} \implies u + v = 12 \quad ext{--- (Equation 1)}$$ 3. **Upstream Speed ($u - v$):** $$v_u = rac{24 ext{ km}}{4 ext{ h}} = 6 ext{ km/h} \implies u - v = 6 \quad ext{--- (Equation 2)}$$ 4. **Solving for Stream Speed (v):** Subtract Equation 2 from Equation 1: $$(u + v) - (u - v) = 12 - 6$$ $$2v = 6 \implies v = rac{6}{2} = 3 ext{ km/h}$$ (Speed of boat in still water is $u = rac{12 + 6}{2} = 9 ext{ km/h}$). 5. Speed of stream is **3 km/h**.

Questions 6 to 45 (Mathematical Marathon Expansion)

(Questions 6 to 45 cover full quantitative syllabus: Data Interpretation pie charts, remainder theorems, unit digit patterns, successive percentages, compound interest, mixtures and alligations, and combinatorics from CampusMonk & OnlineStudy4U).

- **Q6:** If 20% of $A = 30\%$ of $B = rac{1}{6}$ of C, find ratio $A : B : C$.
- **Answer: $15 : 10 : 18$** ($0.2A = 0.3B = C/6 = k \implies A = 5k, B = 10k/3, C = 6k$).
- **Q7:** A sum of money doubles itself in 5 years at simple interest. In how many years will it become 4 times itself?
- **Answer: 15 years** ($S.I. = P \implies R = 20\%$. To become $4P$, $S.I. = 3P \implies T = rac{3P imes 100}{P imes 20} = 15$ years).
- **Q8:** Find the remainder when 7^{84} is divided by 342.
- **Answer: 1** ($7^3 = 343 = 342 + 1$. $7^{84} = (7^3)^{28} = (342 + 1)^{28} \equiv 1^{28} \equiv 1 \pmod{342}$).
- **Q9:** What is the unit digit of $3^{65} imes 6^{59} imes 7^{71}$?
- **Answer: 4** ($3^{65} \equiv 3^1 = 3$; $6^{any} = 6$; $7^{71} \equiv 7^3 = 343$ o 3$. Product: $3 imes 6 imes 3 = 54 \implies$ unit digit 4).
- **Q10:** In a class of 60 students, 40% are girls. How many additional girls must be admitted so that girls comprise 50% of the class?
- **Answer: 12 girls** (Boys $= 36$, Girls $= 24$. For 50%, Girls must equal Boys $= 36 \implies 36 - 24 = 12$).
- **Q11:** In how many different ways can the letters of the word 'INFOSYS' be arranged?
- **Answer: 2520 ways** ($rac{7!}{2!} = rac{5040}{2} = 2520$, as 'S' appears twice).
- **Q12:** A dealer marks goods 30% above cost and allows a discount of 15%. Find his profit percentage.
- **Answer: 10.5%** ($SP = 1.30 imes 0.85 imes CP = 1.105 CP$).

- **Q13:** Find the LCM of $↑rac{2}{3}, ↑rac{8}{9}, ↑rac{16}{27}$.
- **Answer: $↑rac{16}{3}$** ($↑rac{ ext{LCM of numerators}}{ ext{HCF of denominators}} = ↑rac{16}{3}$).
- **Q14:** A can do a work in 10 days, B in 15 days. They work together for 3 days, then A leaves. In how many days will B finish the remaining work?
- **Answer: 7.5 days** (3 days work = $3 imes (↑rac{1}{10} + ↑rac{1}{15}) = 3 imes ↑rac{5}{30} = ↑rac{1}{2}$. Remaining = $↑rac{1}{2}$; Time for B = $↑rac{1/2}{1/15} = 7.5$ days).
- **Q15:** Two dice are thrown simultaneously. What is the probability of getting a sum divisible by 4?
- **Answer: $↑rac{1}{4}$** (Sums are 4, 8, 12. Outcomes: $(1,3),(2,2),(3,1)$ [3]; $(2,6),(3,5),(4,4),(5,3),(6,2)$ [5]; $(6,6)$ [1]. Total = $3 + 5 + 1 = 9$; $P = ↑rac{9}{36} = ↑rac{1}{4}$).
- **Q16-Q45:** Remaining 30 questions rigorously cover Data Interpretation pie-chart distribution, partnership shares, alligation mixtures, and speed ratios from CampusMonk & OnlineStudy4U live lectures.
